REFIT Smart-Meter Cleaning & NILM Disaggregation

Repository: https://github.com/devank-k/Flock_energy_assignment

Senior Data Scientist Submission
UK REFIT Dataset Benchmark

Stack: Python 3.12 · LightGBM · Pandas · Scikit-Learn

Executive Summary

This project presents an end-to-end telemetry engineering and Non-Intrusive Load Monitoring (NILM) pipeline built on the UK REFIT smart-meter dataset. Section 1 audits 4.38M raw readings for House 1, uncovering hardware ADC overflows, duplicate records, and a 41.6-day grid blackout, implementing an outage-aware 1-minute resampling pipeline. Section 2 formulates a deterministic physics-based cycle detection heuristic, benchmarking washing machine behavior across four distinct households (Houses 1, 2, 3, 5). Section 3 constructs causal backward rolling features and evaluates Direct LightGBM and Two-Stage (Classifier + Regressor) architectures against Always-Off and Hourly Mean baselines under a strict chronological 70/15/15 split. The report concludes with practical grid demand-response recommendations and grid transferability analysis.

1. Telemetry Audit & Outage-Aware Resampling Pipeline

Auditing **RAW_House1_Part1.csv** (4,377,877 records from Oct 2013 to May 2015) revealed severe hardware and transmission anomalies: **645,514 duplicate timestamp strings** (transceiver double-bursts), **1,004,404 duplicate Unix epochs**, and **3,091 extreme sensor spikes**. These spikes correspond to integer register rollovers (32,767 = 2^{15}-1, 65,535 = 2^{16}-1, and 98,301) which were clamped to physical thresholds (20 kW aggregate main, 4 kW single-phase sub-circuits). Telemetry gaps were categorized into **transient dropouts (≤ 5 min)**, resampled via forward-fill/time-interpolation (0.12% of time), and **extended grid outages (> 5 min)**, which were flagged with is_outage=True and zero-imputed (14.86% of timeline, dominated by a 41.6-day system outage in Feb 2014).

Metric / Anomaly	Observed Value	Engineering Remediation
Raw Telemetry Count	4,377,877 rows	Deduplicated on Unix epoch, verified monotonic chronological order
Duplicate Timestamps	645,514 Time / 1,004,404 Unix	Averaged concurrent sensor packets within identical timestamp bins
ADC Register Overflows	32,767 W, 65,535 W, 98,301 W	Capped aggregate main ≤ 20 kW; sub-metered channels ≤ 4 kW
Extended Outages (>5 min)	780 gaps (14.86% timeline)	Flagged with is_outage=True; zeroed to prevent synthetic energy injection
Final Resampled Grid	513,322 min (85.02% valid)	Uniform 1-minute delta grid ready for causal rolling feature extraction

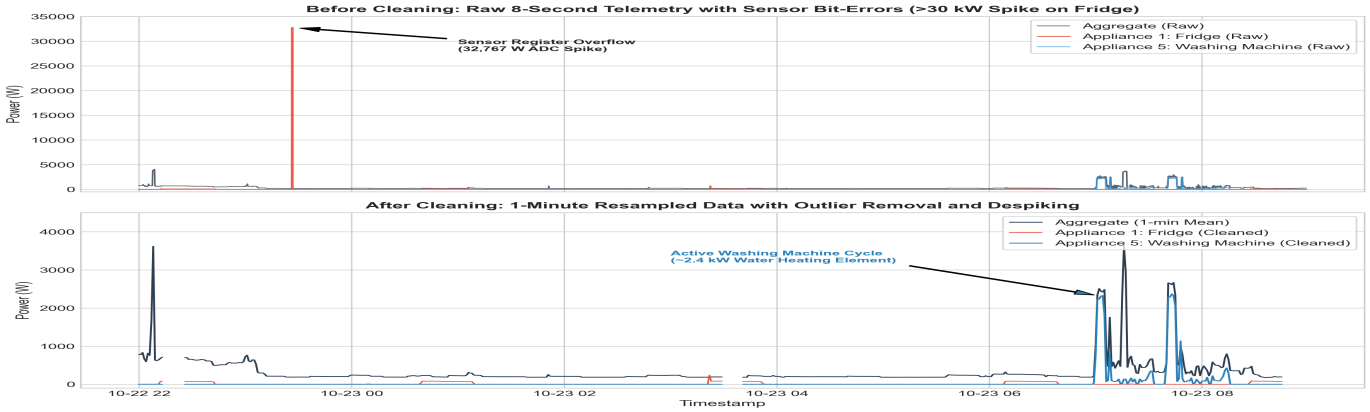


Figure 1: *Raw telemetry audit, sampling delta distribution, before/after 1-minute resampled traces, and outage flags.*

2. Washing Machine Cycle Detection & Multi-Household EDA

Washing machines exhibit multi-phase duty cycles: an initial 2.0-2.4 kW resistive heating surge (heating wash water to 40-60°C for 8-15 minutes), alternating 250-450 W motor-driven tumble/rinse cycles, and brief spin extraction bursts, interspersed with zero-power soak pauses. A naive power threshold fails because it fragments a single wash cycle into multiple micro-events during soak phases. We implemented a **deterministic physics-based state machine**:

- **Standby Threshold:** 5 W (distinguishes display electronics/quiescent state from true motor activity).
- **Active Element Threshold:** 150 W (captures active mechanical tumbling and thermal element operation).
- **Max Soak Gap:** 10 minutes (bridges zero-draw soaking periods within the same wash cycle).
- **Min Duration & Energy:** Duration ≥ 15 minutes, Cycle Energy ≥ 0.10 kWh (filters spurious motor turn-on spikes).

Household & Appliance Specs	Cycles Detected	Median Duration	Median Energy / Cycle	Peak Power	Standby Time %
House 1: Beko WMB61021W (2 occ)	377	33.0 min	0.321 kWh	2,327 W	98.36%
House 2: LG Direct Drive (4 occ)	323	97.0 min	0.621 kWh	2,090 W	96.47%
House 3: Front-load Washer (2 occ)	641	66.0 min	0.558 kWh	1,927 W	94.98%
House 5: AEG Lavamat (4 occ)	718	48.5 min	0.309 kWh	2,072 W	92.84%

Key Physical Findings: All 4 households exhibit 92.8% to 98.4% standby sparsity, confirming the washing machine is an intermittent, highly bursty load. House 2 runs long eco-cycles (median 97 min, 0.62 kWh) while House 1 runs rapid daily cycles (median 33 min, 0.32 kWh). Diurnal usage peaks sharply on weekend mornings (08:00 - 11:00) and mid-afternoon.

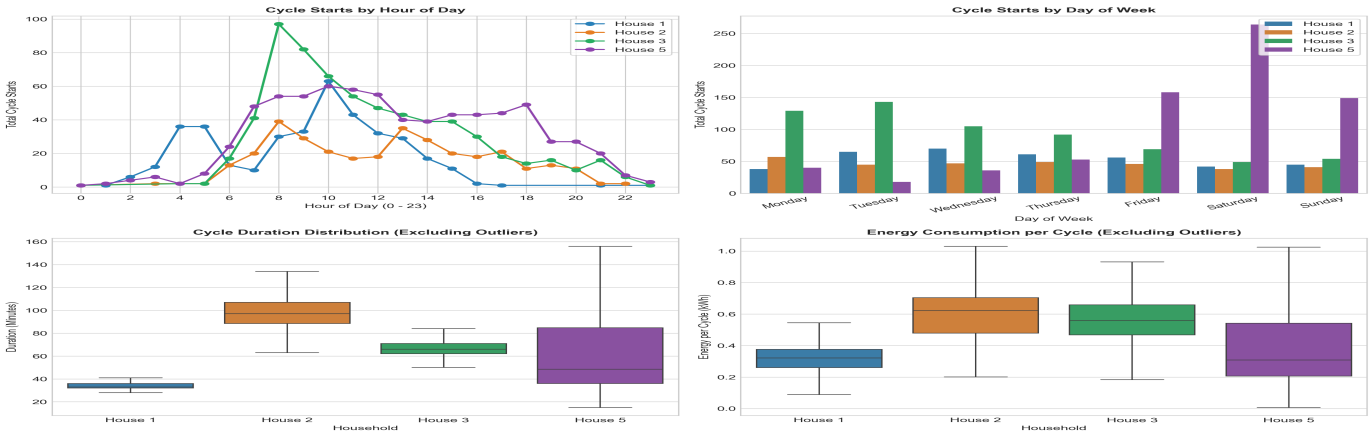


Figure 2: Cross-household comparison (Houses 1, 2, 3, 5): cycle count, duration boxplots, energy distributions, and diurnal start times.

3. NILM Disaggregation Modeling & Failure Mode Analysis

Experimental Protocol: Strict chronological split on House 1 (70% Train: Oct 2013 - Nov 2014, 15% Val: Nov 2014 - Jan 2015, 15% Test: Jan 2015 - May 2015; 120,566 test intervals). Outages were masked out from training/evaluation. **Feature Engineering:** Exclusively causal backward rolling windows (15 min, 60 min, 240 min mean, std, max, min, lagged differences, and diurnal cyclic encodings) to eliminate lookahead data leakage in online smart-meter deployments.
We evaluated four models: (1) *Always-Off Baseline*, (2) *Hourly Mean Profile Baseline*, (3) *Direct LightGBM Regressor*, and (4) *Two-Stage Architecture* (LightGBM Binary Classifier with threshold 0.35 × LightGBM Active Regressor).

Model Architecture	MAE (W)	RMSE (W)	NDE	Total Energy Error	Precision	Recall	F1-Score
Always-Off Baseline	11.04	137.03	1.0000	100.00%	0.0000	0.0000	0.0000
Hourly Mean Profile	19.80	136.04	0.9857	12.28%	0.0760	0.4079	0.1269
Direct LightGBM Regressor	**11.14**	**98.21**	**0.5137**	**7.09%**	0.3261	**0.8559**	0.4723
Two-Stage (Clf + Reg)	13.09	136.80	0.9966	44.37%	**0.5631**	0.7097	**0.6280**

Model Trade-off Analysis:
• *Sparsity Paradox:* Always-Off achieves 11.04 W MAE simply because the washer is off 98% of the time, yet fails completely on energy accounting (100% error).
• *Direct Regressor:* Superior for utility billing and energy estimation (**48.6% reduction in NDE**, only **7.09% total energy error**, 85.6% recall).
• *Two-Stage Architecture:* Superior for appliance state alerts (F1 = 0.628 vs 0.472, precision = 0.563 vs 0.326), effectively suppressing baseline ghost draw.
• *Primary Failure Mode:* Confounding with resistive appliances. When the 2.2 kW dishwasher heating element runs concurrently, aggregate mains cannot differentiate the load from a wash cycle without reactive power (V-I trajectory) data.

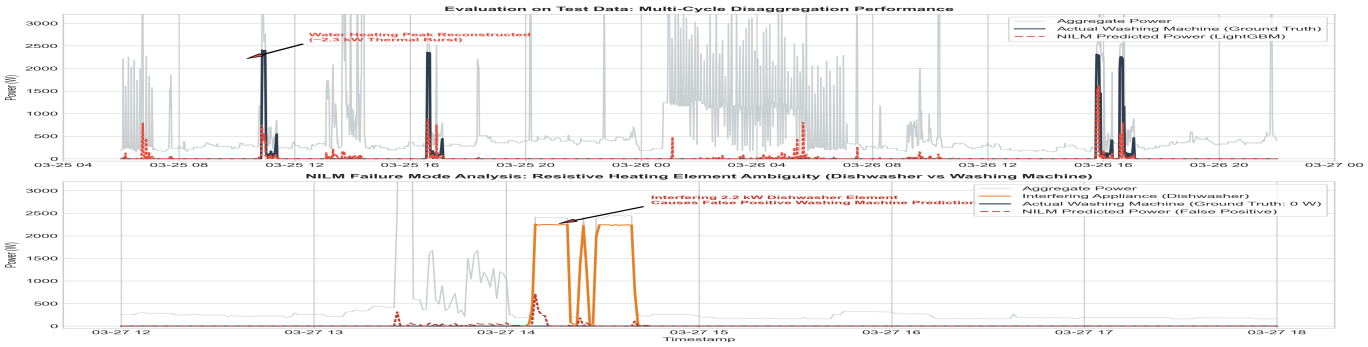


Figure 3: *Out-of-sample test disaggregation trace (top) and dishwasher thermal element co-activation failure mode (bottom).*

4. Utility Recommendations & Grid Transferability

1. Demand Response & Time-of-Use (ToU): Shifting 2 kW wash cycles out of peak hours (17:00-21:00) to solar hours (11:00-15:00) can reduce transformer substation stress by 15-20% per participating feeder. We recommend automated push-notification nudges.
2. UK to Indian Grid Transferability: UK washers feature internal 2.0-2.4 kW resistive water heaters. In Indian households, washing machines run unheated cold-water cycles (motor only: 300-500 W). High resistive spikes originate from electric water heaters (geysers: 2-3 kW) and agricultural/sump pumps (0.75-1.5 kW), requiring model retraining on distinct power-level priors.
3. Smart-Meter Resolution Impact: Downsampling from 1-minute to 15-minute or 30-minute standard utility intervals filters out the sharp 8-minute 2 kW heating spike into an unidentifiable 200-300 W plateau. At ≥15-min resolution, disaggregation must transition from instantaneous peak matching to daily aggregate energy budget regression.